



Sardar Patel Institute of Technology
Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India
(An Empowered Autonomous College Affiliated to University of Mumbai)
Department of Computer Engineering

Vehicle Rental Management System – Data Flow Diagram

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Aim:

To draw the Level 0, Level 1 and Level 2 Data Flow Diagrams for the Vehicle Rental Management System using Gane and Sarson notation.

Implementation:

The data flow diagrams were derived from the Software Requirements Specification and kept consistent with the use case, class, sequence and activity diagrams of the earlier experiments. The context diagram (Level 0) shows the system as a single process exchanging data with its external entities. Level 1 explodes that process into eight numbered processes and the data stores they read and write. Level 2 explodes three of those processes – Manage Booking, Handle Vehicle Return and Settlement, and Manage Vehicle Maintenance – into their component processes. Process numbering is hierarchical, so process 3 at Level 1 becomes 3.1 to 3.7 at Level 2, and the data flows crossing each boundary balance with the level above it.

Gane and Sarson Notation:

- Process – a rounded rectangle divided by a horizontal line. The upper band carries the process number and the lower area carries the process name.
- Data store – an open-ended rectangle with a shaded left cell carrying the store identifier (D1, D2, ...) and the store name alongside it.
- External entity – a plain rectangle naming a source or destination of data that lies outside the system boundary.
- Data flow – a labeled arrow showing data moving into or out of a process, data store or external entity.

Rules followed:

- Every process has at least one input and one output data flow.
- Data flows never run directly between two external entities, or between two data stores; they always pass through a process.
- Every data store is both read and written somewhere in the diagram set.
- Each data flow is labeled with the data it carries, not with the action performed.
- Child diagrams balance with their parent: the flows entering and leaving a Level 2 diagram are the same flows attached to that process at Level 1.



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Level 0 – Context Diagram

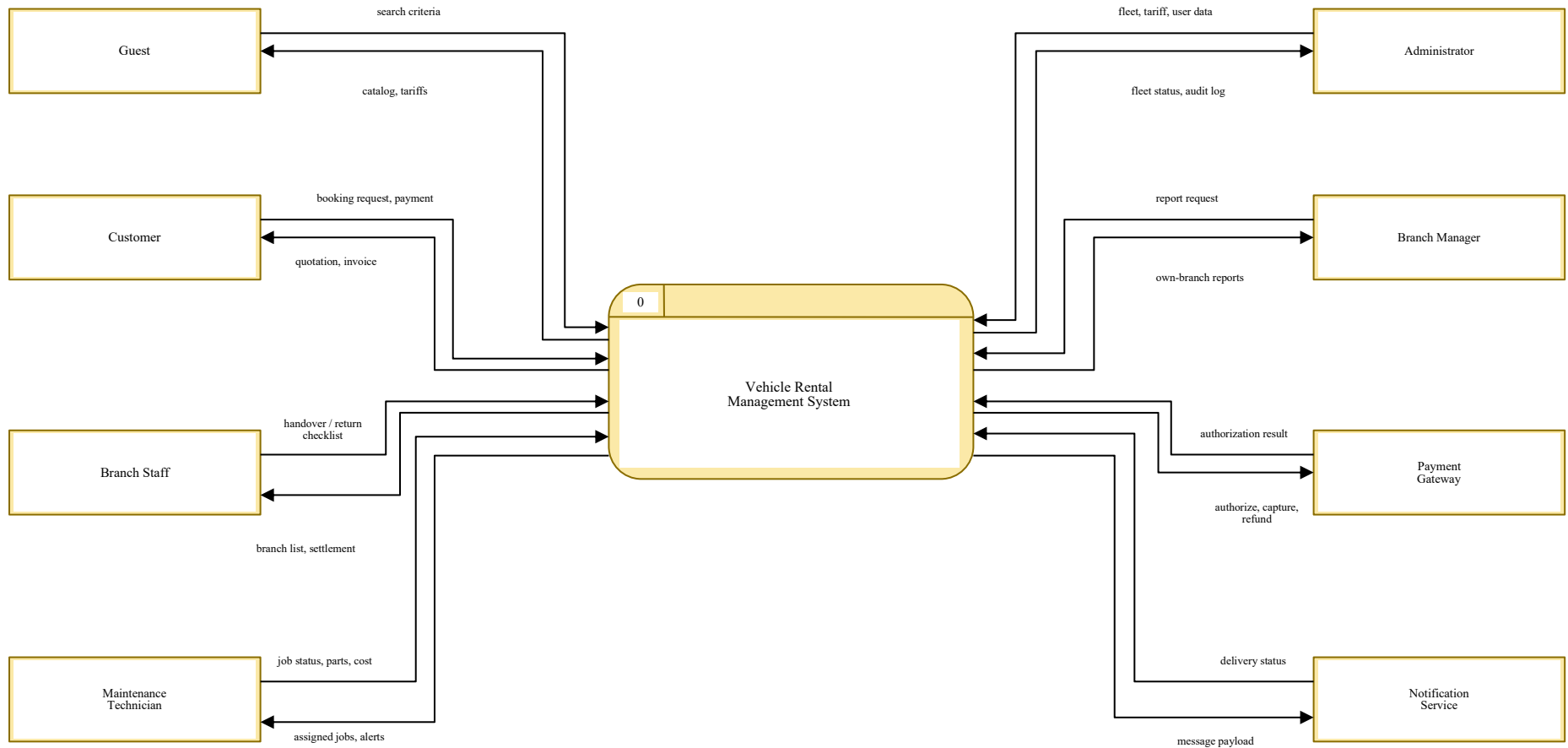


Figure 1. Level 0 (context) data flow diagram.



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Level 1 – System Decomposition

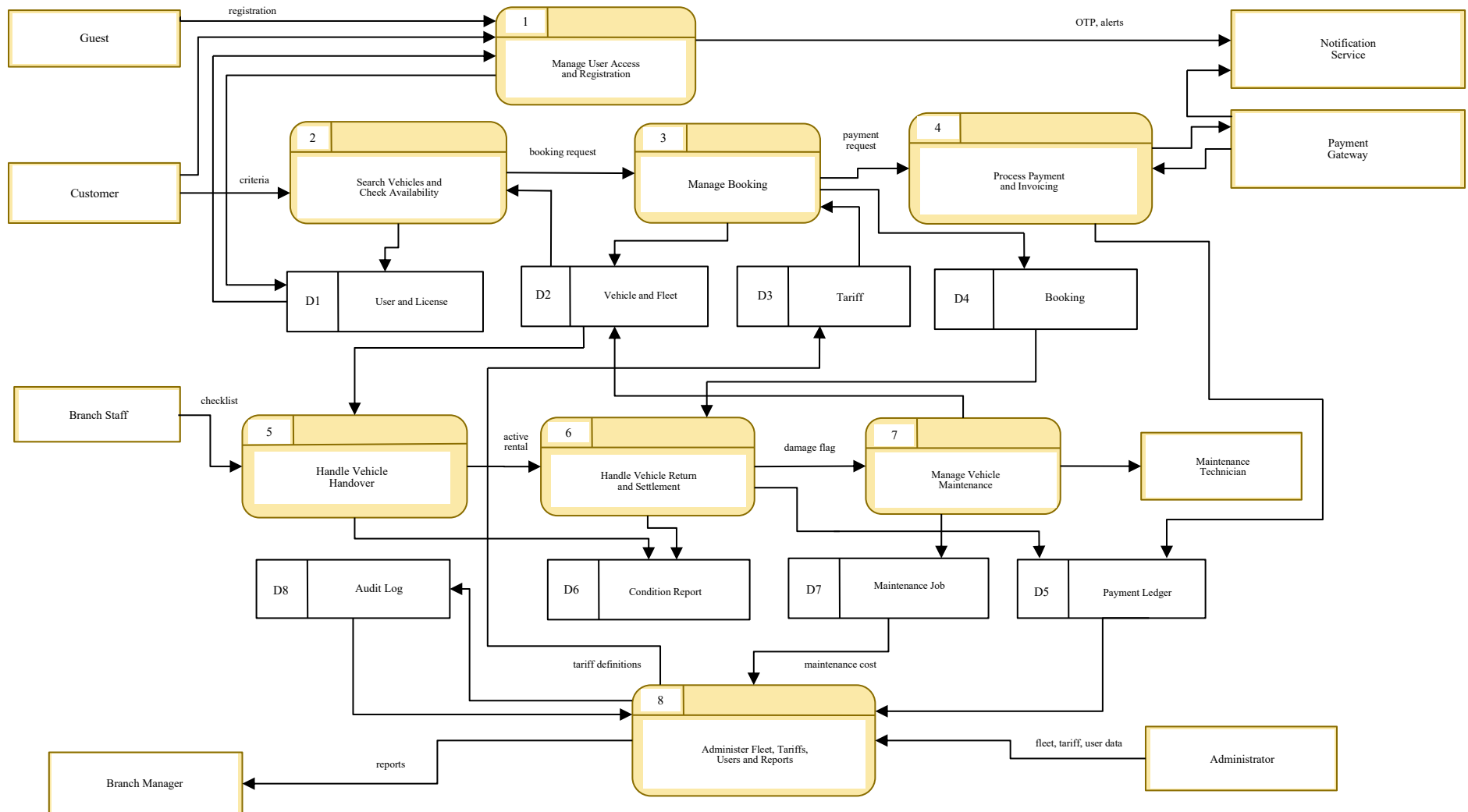


Figure 2. Level 1 data flow diagram.



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Level 2 – Process 3: Manage Booking

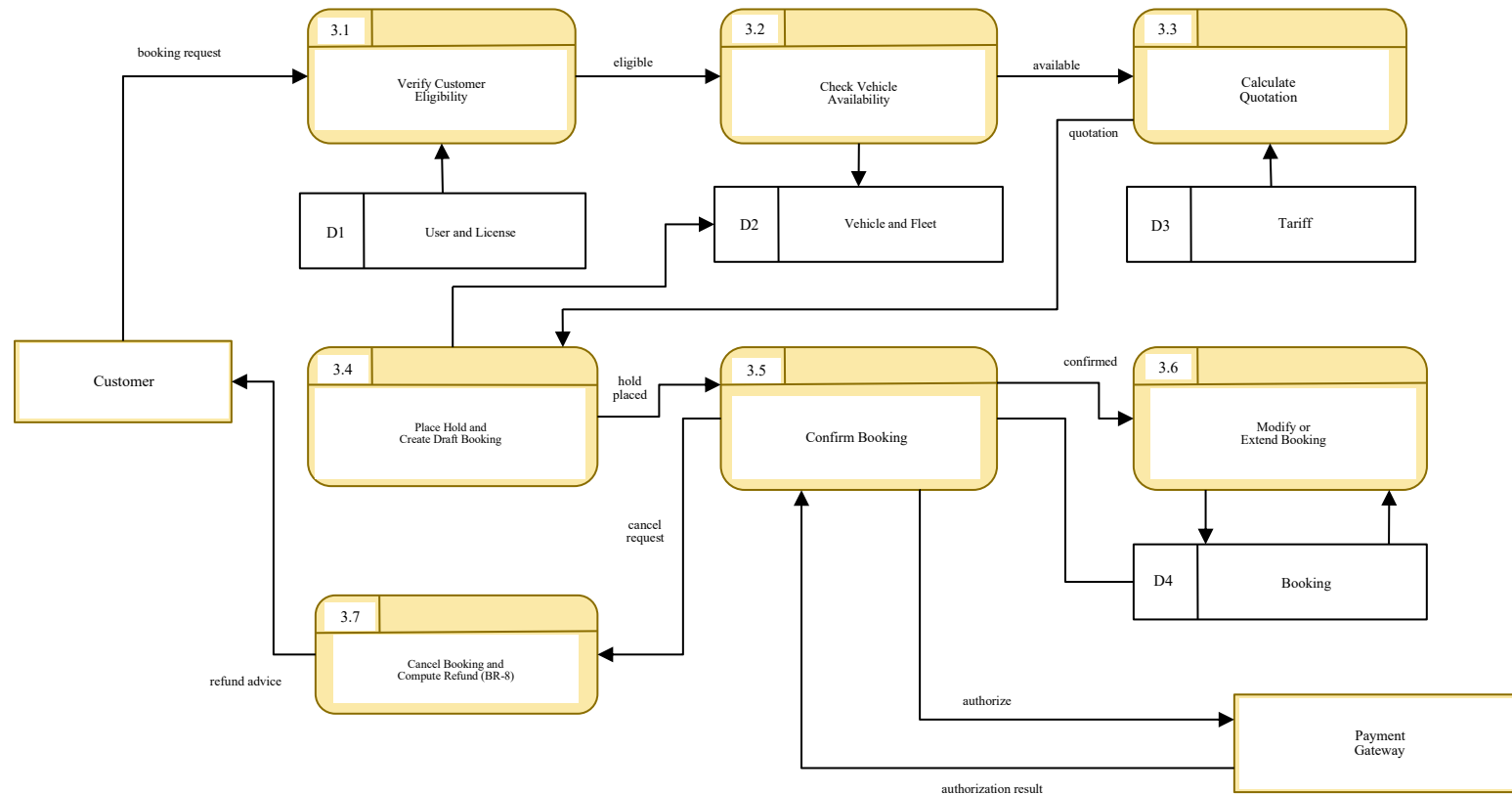


Figure 3. Level 2 explosion of process 3.



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Level 2 – Process 6: Handle Vehicle Return and Settlement

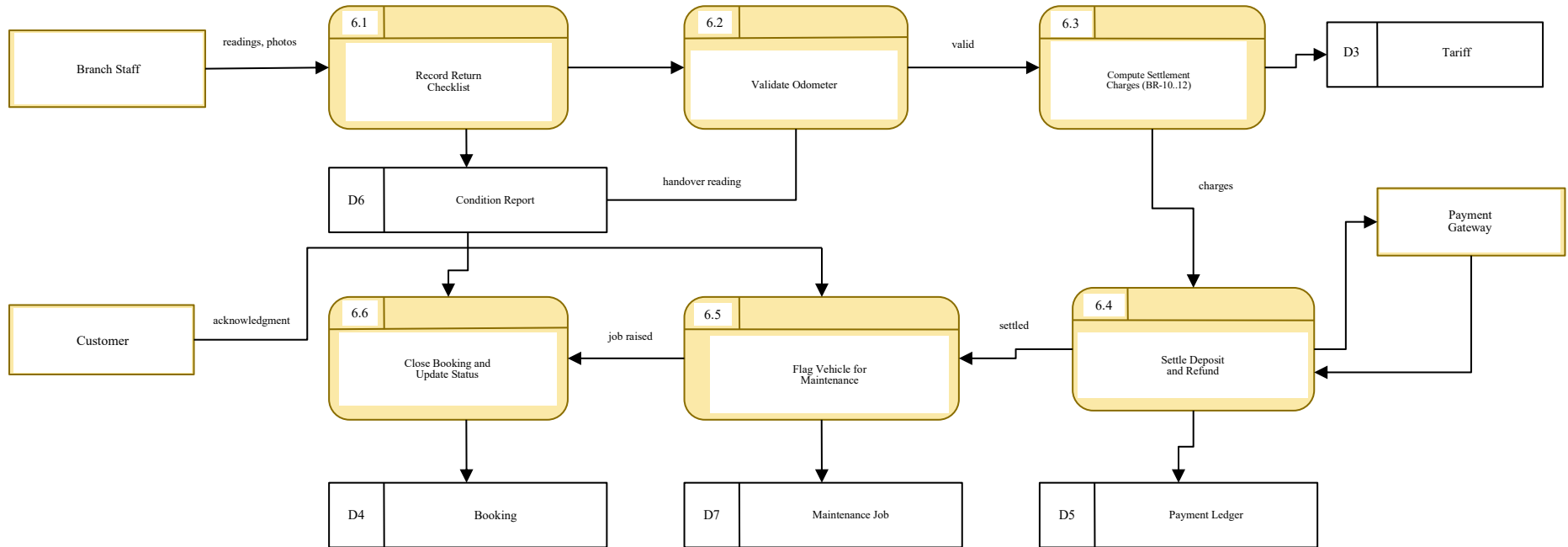


Figure 4. Level 2 explosion of process 6.



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Level 2 – Process 7: Manage Vehicle Maintenance

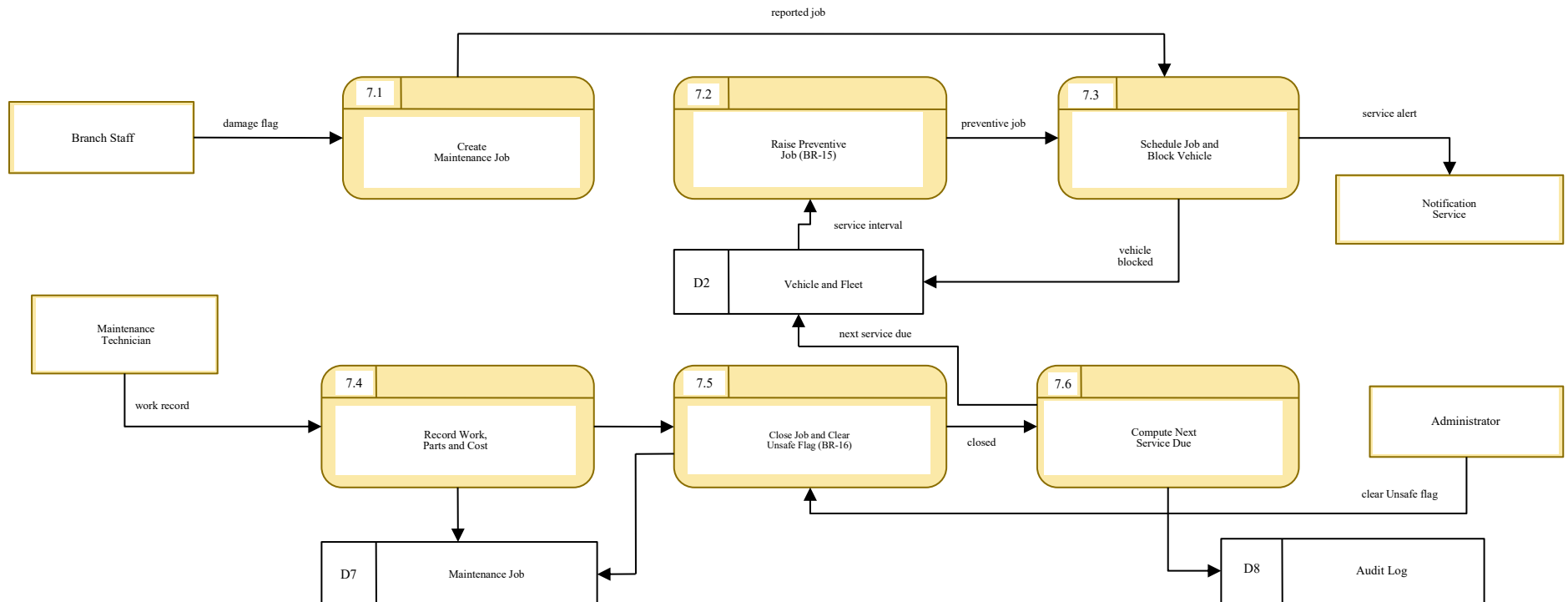


Figure 5. Level 2 explosion of process 7.

Conclusion:

By performing this experiment we learned how to model the flow of data through the Vehicle Rental Management System using Gane and Sarson notation. The context diagram fixed the system boundary and identified the eight external entities the system exchanges data with. The Level 1 diagram decomposed the system into eight processes and eight data stores, showing which process writes each store and which reads it. The Level 2 diagrams explained the three most complex processes in further detail, keeping the hierarchical numbering and balancing the flows with the parent diagram. We also learned the rules that make a data flow diagram valid, in particular that data can never move directly between two external entities or between two data stores, and that every process must both consume and produce data.